The Helmholtz Knowledge Graph: Building a semantic interoperability layer across the Helmholtz Association

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Abstract

Large, distributed research organizations face ongoing challenges in ensuring that their research outputs are findable, interoperable, and reusable across institutional boundaries. The Helmholtz Association comprises 18 independently operating research centers across Germany, resulting in a highly heterogeneous digital asset landscape with respect to formats, metadata schemas, record consistency, and hosting locations and operations. Metadata describing publications, datasets, software, and other entities related to scientific research is spread across numerous institutional repositories, formats, and interfaces. This creates barriers to cross-institutional data discovery and integration, preventing researchers from easily finding and building upon relevant work across Helmholtz centers. As a result, data exchange based on commonly agreed principles within a FAIR Helmholtz Data Space is limited, hindering the full realisation of the data's value. This work addresses these challenges by presenting an operational semantic infrastructure that integrates and harmonizes decentralized research metadata across the Helmholtz Association.

Keywords

Knowledge Graph, FAIR Data, Metadata Integration, Semantic Interoperability, Research Infrastructure, RDF, SPARQL, OpenSearch, Helmholtz Association

1. Introduction

The Helmholtz Metadata Collaboration (HMC)[1] is taking on this challenge by developing the Helmholtz Knowledge Graph (Helmholtz KG) [2] as a lightweight interoperability layer and semantic harmonization target, that connects metadata of Helmholtz digital assets, which are stored in a decentralized manner.

With the KG, HMC envisioned (1) providing better cross-organizational access to Helmholtz's (meta)data and information assets on an upper semantic level, (2) harmonizing and optimizing the related metadata across the association, and (3) forming a basis from which the semantic quality and the depths of metadata descriptions is improved and extended into domain and application levels.

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The Knowledge Graph infrastructure processes metadata in a neutral and transparent way, reflecting what is provided by source systems and how it aligns with shared schemas. As a result, observed issues such as missing links or duplication arise from interactions between source metadata and pipeline processing. Understanding these interactions is key to improving metadata quality in exchange with our data providers.

2. Approach

The Helmholtz Knowledge Graph is more than a graph database or RDF triple store; it is a semantic infrastructure designed to observe, analyze, and understand metadata across distributed scientific systems within the Helmholtz ecosystem. This is achieved through coordinated harvesting, harmonization, and query components that enable systematic assessment of metadata and active endpoints rather than mere storage of information.

The current system, publicly available through the Helmholtz KG [3] repository, implements a modular data pipeline consisting of:

1. A set of harvester libraries for OAI-PMH, DataCite APIs, sitemaps guided harvesting of HTML embedded metadata, and custom de-facto standardized endpoints;
2. A storage API layer that provides RESTful access to the harvested raw data;
3. A mapping and transformation layer with an data model that leverages schema.org fields with few semantic extensions, used to normalize heterogeneous metadata into structured, schema-aligned entities;
4. A graph database that stores mapped and aligned information as Resource Description Framework triples.
5. A set of utilities that orchestrate communication between system components, including a wrapper for coordinating data ingestion and import processes, with a parallel OpenSearch layer providing fast text-based discovery through indexed metadata fields.;
6. Dual access interfaces consisting of an SPARQL endpoint for graph queries and an User Interface (UI) for web-based access and keyword-based exploration.

This hybrid architecture (RDF triples vs data index) provides complementary capabilities across the system stack: The graph database component handles semantic relationships, entity linking, persistent identifier collection and cross-referencing (DOI, ORCID, ROR), and SPARQL-based queries, while OpenSearch enables fast keyword search, faceting, and ranking for discovery workflows. This dual-storage approach enables complex queries via the SPARQL endpoint, as well as user-friendly filtering and free-text search through the UI based on the OpenSearch index.

3. Operational Characteristics

The system's operational capabilities include incremental harvesting with configurable date ranges, multi-threaded endpoint execution for parallel data collection, and batch processing optimizations that enable scalable integration across many heterogeneous sources. In the current development stage, metadata from multiple Helmholtz centers undergoes regular harvesting cycles, schema.org-based normalization, cross-institutional entity linking, and integration into a unified knowledge graph that spans research domains and institutional boundaries. The modular architecture ensures database-agnostic deployment flexibility, supporting various graph database backends while maintaining consistent SPARQL query interfaces and semantic data models throughout the pipeline.

Declaration on Generative AI

During the preparation of this work, the author(s) used GPT-5.2 in order to grammar and spelling check and to assist in text condensation. After using this tool, the author(s) reviewed and edited the content

as needed and take(s) full responsibility for the publication's content.

References

[1] Helmholtz Metadata Collaboration, (HMC), n.d. URL: https://helmholtz-metadaten.de.

[2] J. Bröder, O. Brenndike-Mannix, G. Preuß, S. Fathalla, F. D'Mello, M. Kubin, V. Hofmann, The Helmholtz Knowledge Graph: Driving the Transition Towards a FAIR Data Ecosystem in the Helmholtz Association, in: Proceedings of the Extended Semantic Web Conference (ESWC), 2024. URL: https://2024.eswc-conferences.org/wp-content/uploads/2024/05/77770177.pdf, conference paper.

[3] Helmholtz Knowledge Graph (HKG) - Public Development Repository, n.d. URL: https://codebase.helmholtz.cloud/hmc/hmc-public/unhide/development.